

Toward a precise theory of imprecise representations

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Abstract

Models in cognitive science typically explain behavioral patterns by positing the kinds of computations, often presumed suboptimal, that the brain might implement. Implicitly, these computations are assumed to operate on representations that faithfully reflect the relevant information. In the brain, however, representations are imprecise, with broad repercussions for many cognitive processes. A quantitative theory of imprecise representations is therefore a necessary foundation for understanding cognition. Here we develop the basis for such a theory, and we draw out some of its implications. From a single principle of endogenous efficient coding, we derive six quantitative laws of psychophysics, including Weber's law, scaling laws in visual working memory, and Shepard's universal law of generalization. These results, supported by neural evidence, suggest that the brain is able to finely adapt its representational precision to the statistics and demands of the current task. Context thus shapes internal representations, with profound and predictable implications for perception and decision-making.

A venerable theoretical tradition in cognitive science treats the mind as an information processing system that performs computations over internal representations. Cognitive scientists thus seek to infer people's computational strategies in various tasks, such as: how they search through problem space in problem-solving tasks; how they update expected values, in reinforcement-learning scenarios; how they weigh gains and losses and distort probabilities in decisions under uncertainty; and how they implement belief updates in inference tasks. Many theories propose explanations regarding how people manipulate the information presented to them, implicitly assuming that this information is perfectly represented. Consider, for example, risky choices between lotteries. Here the information to be processed is minimal: just a few unambiguous numbers (e.g., '\$1 with probability 0.5'). In traditional models, these numbers undergo in the mind of the decision-maker a series of transformations (e.g., utility function, probability-weighting function, and comparison) that determine the decision. Behavioral patterns are then explained as resulting from the properties of these computations.

We propose instead that the information presented in most tasks is not precisely represented in the brain, with

far-reaching consequences for decision-making. Numbers, to continue our example, are mapped to internal representations that are imprecise (in ways we detail below), such that the numbers cannot be unambiguously recovered from them. Yet it is on these 'fuzzy' representations that decisions are made. In risky choices, behavioral and neurophysiological results demonstrate that cognitive imprecision accounts for choice behavior¹⁻⁷, in departure from traditional theories of risk attitudes. This suggests that before studying complex human decisions, it is useful to first understand how the brain represents the elements that inform them.

A theory of representations would provide sound foundations for cognitive models of decision, inference, learning, memory, etc. We argue that such a theory is within reach, at least for magnitudes, such as dollar amounts and probabilities as in the example above, but also other kinds such as durations, distances, temperatures, ratings, and so on, and potentially more abstract concepts that lend themselves to comparisons (e.g., "better than", "tastier than", etc.). Such scalar representations are fundamental to many perceptual and decision tasks (including many economic decisions: accordingly, behavioral economists have shown growing interest in various kinds of imperfect information processing⁸⁻¹⁹). We present the theoretical efforts undertaken in this direction, the experimental results supporting them, and the predictions that remain to be tested.

The theory we present, in its current state, is a careful formalization of the idea that the brain forms optimal representations, subject to the constraints under which it operates. This perspective is related to efficient coding in neuroscience²⁰⁻³¹, computational or resource rationality in cognitive science³²⁻³⁵, and rational inattention in economics⁸⁻¹¹. Each of these frameworks proposes that the brain optimizes some objective under some constraint; but none seem to agree on the objective nor on the constraint³⁶. Yet these elements determine representations in these models, and should therefore be clearly identified. A crucial ingredient in the theory we propose is that representational precision comes at a cost. Unlike a fixed capacity constraint, a cost allows the brain to finely choose how precisely to represent information, as a function, for instance, of task demands. This

flexibility has important behavioral consequences, which we present below. The idea is closely related to notions of cost of mental effort, cost of cognitive control, and attentional cost, developed in different strands of the literature^{37–43}.

The theoretical framework predicts laws of psychophysics, including long-established laws (e.g., Weber’s law), but also new laws that we show are verified in data, and new laws that have yet to be tested. In particular, it predicts non-trivial scaling laws for the variability of percepts and for biases. We emphasize that all the behavioral effects we present here are *derived* from first principles (in fact, from a single equation). The goal is to unearth these fundamentals, rather than to exhibit behavioral curiosities. We also point to neural evidence that directly supports the theory.

An important insight that emerges from this body of work is that representations are exquisitely sensitive to the context in which they are needed. For instance, the representation of a magnitude is different depending on whether it is used for estimating the magnitude, or for comparing it with another magnitude. In other words, representations depend on what they are used for. In addition, they depend on the statistics of the represented information (i.e., the prior). The argument is not only that context influences *decisions* (as Bayesian decision theory predicts), but that it determines *representations*, on which reasoning and decision-making operate. This sensitivity has important implications for the study of human cognition. Careful attention should be paid to the precise conditions under which behavior is elicited; these conditions shape representations, and therefore they shape cognition, and behavior.

A theory of magnitude representations: endogenous efficient coding

Given an observer presented with a magnitude x , our objective is to formulate some description of the representation r elicited in the observer’s brain. The theory we present instantiates the principle of efficient coding, i.e., the notion that the brain chooses optimal representations, under constraints. This is formalized as an optimization problem. We then show that from this single optimization problem follow six laws of psychophysics that are widely supported in data. Throughout, we assume that the observer decodes noisy representations optimally, as in the ‘Bayesian efficient-coding’ framework⁴⁴.

Characterizing representations To define the optimization problem, we must specify what is optimized over. Here this is the ‘encoding’, i.e., the distribution $p(r|x)$ of the representation r conditional on the magnitude x . In economics, this would be called the ‘signal structure’. In neuroscience, this means for instance choosing neuronal tuning curves, i.e., specifying how the (stochastic) neural activity encodes the magnitude, a complex and often ill-defined problem. Instead we characterize the encoding by its *Fisher information*, $I(x)$, which quantifies its precision (Box 1). Different neural implementations may induce the same Fisher information,

and are therefore equivalent for our purposes: working with $I(x)$ allows us to abstract away from idiosyncratic features of the code (e.g., the precise parametrization of tuning curves) and to formulate general principles of neural coding.

This provides a compact description of representations while retaining direct links to behavior: in the small-noise regime, the standard deviation of estimates and the discrimination threshold, or just-noticeable difference (JND), are both approximated (up to a constant) by $\sigma(x) = 1/\sqrt{I(x)}$, and it also defines an important component of the bias (see Box 1 and Eq. 13). We make one structural assumption on the encoding: we assume that it results from the accumulation of n independent, identically-distributed signals (‘samples’), each with ‘one-sample’ Fisher information $I_1(x)$. The resulting total Fisher information is the sum $I(x) = nI_1(x)$. Increasing n improves the overall precision of the encoding, while by choosing the function $I_1(x)$, under some constraint detailed below, one can choose to be more precise about some stimuli than about others.

Loss function The choice of representation is guided by a loss function, which specifies how strongly a given response is penalized in a given situation—for instance, the penalty incurred when estimating a magnitude x as $\hat{x}$. For our purposes, we use the corresponding *expected* loss, obtained by averaging over situations and responses.

We posit that representations are chosen to optimize what the observer is currently trying to do. This may seem like an obvious choice, but it has profound implications, and in many models it is not the chosen objective. Many efficient-coding models assume that the brain maximizes the mutual information (MI) between stimulus and representation. This is not an objective that corresponds in any obvious way to a goal one might have in everyday life; but it may be a sensible general-purpose objective, appropriate in early sensory systems²⁷. However, a fixed objective such as MI implies that representations do not change when the observer’s goal changes; but experimental evidence suggests that they do (see below).

A more flexible hypothesis is that representations are adapted to the task at hand. Myriad tasks are conceivable; here we model a subset of them (Box 2). Each task induces an expected loss, which can be expressed as a functional of the Fisher information, $L_{task}[I]$. These tasks are naturally organized using two indices. The first index, $m \in \{1, 2\}$, distinguishes *estimation* and *choice tasks*: $m = 1$ in estimation tasks, where the goal is to estimate the stimulus magnitude x ; $m = 2$ in choice tasks, where the goal is to choose the stimulus $x \in \{x_1, x_2\}$ with greater expected utility (often greater magnitude, in psychophysical experiments). The second index, $p > 0$, determines how errors are penalized; in estimation tasks, it indicates p^{th} -power errors (see Box 2; below, unless otherwise specified, we consider squared errors, i.e., $p = 2$). The expected loss also depends on the stimulus statistics, i.e., the prior. We denote by $L_{f,m,p}[I]$ the expected loss associated with a task characterized by m and p , in the context of a prior f .

Box 1: A primer on Fisher information

Perceptual neuroscience suggests that representations materialize in the activity of sensory neurons, whose stochastic firing patterns depend on the stimulus, thereby thought to be *encoded*. The precision of the representation is therefore the precision of this neural code. Two stimuli will easily be distinguished if the neural patterns they elicit are different. The Fisher information, $I(x)$, formalizes this idea: it is a function of the derivative of the probability distribution characterizing the neural activity, $p(r|x)$, with respect to the encoded stimulus, x . Thus it measures the sensitivity of the encoding to x . Formally,

$$I(x) = \mathbb{E} \left[\left(\frac{d}{dx} \log p(r|x) \right)^2 \mid x \right].$$

For independent signals the Fisher information simply sums, i.e., n signals with Fisher information $I_1(x)$ together have total Fisher information $I(x) = nI_1(x)$. We now detail important and useful properties of this quantity.

Measure of imprecision Consider a stimulus x , encoded in n stochastic signals collectively denoted by r , with total Fisher information $I(x)$. For the Bayesian-mean estimator, $\hat{x}(r) = \mathbb{E}[x|r]$, (i) the variance of the estimator, $\text{Var}[\hat{x}|x]$, (ii) its mean squared-error, $\mathbb{E}[(\hat{x} - x)^2|x]$, and (iii) the variance of the Bayesian posterior, $\text{Var}[x|r]$, are all approximately (with large n and under regularity conditions^{30,45})

$$\sigma^2(x) = \frac{1}{I(x)}.$$

We clarify that $\sigma(x)$ is not the standard deviation of x : rather, it is the imprecision of the estimator when the stimulus is x , where ‘imprecision’ can be understood in any of the senses (i-iii) above. (We focus on the first two, but experimental measures of the third one, i.e., of observers’ uncertainty, indeed correlate with the first two⁴⁶.) This property is useful to characterize the loss in an estimation task (e.g., quadratic loss; see Box 2) as well as the behavioral data: $\sigma(x)$ approximates the standard deviation of estimates, in estimation tasks, and it can be shown that it is proportional to the discrimination threshold (or JND), in discrimination tasks.

Encoding effect on the bias Another important property is that the variation of the Fisher information across stimuli affects the bias of the estimator, by pulling it towards the *less-precisely* encoded stimuli (Eq. 13). We explain in Fig. 1 (this box) the intuition for this ‘encoding effect’ on the bias.

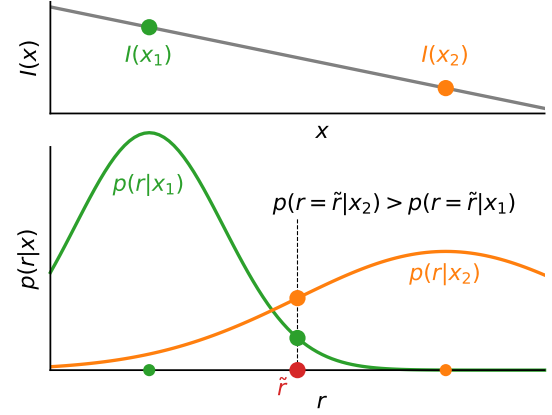


Fig. 1: Encoding effect: a bias toward less-precisely encoded stimuli. In this illustration, the Fisher information for x_1 is greater than for x_2 (top panel). Thus the representation r is more precise for x_1 than for x_2 (narrower distribution $p(r|x)$; bottom panel). Given a realization $\tilde{r}$ of the representation that could result from either x_1 or x_2 , the likelihood of x_2 is greater than that of x_1 , (i.e., $p(r = \tilde{r}|x_2) > p(r = \tilde{r}|x_1)$), due to the stronger specificity of the encoding for x_1 . This greater likelihood induces a bias toward x_2 over x_1 .

Reparameterization and capacity constraint Finally, given a differentiable transformation $z = \varphi(x)$, the Fisher information with respect to z , $I_z(z)$, is related to the Fisher information with respect to x , $I_x(x)$, as

$$I_x(x) = I_z(\varphi(x))(\varphi'(x))^2.$$

Informally, the sensitivity to x is equal to the sensitivity to z times the sensitivity of z to x (chain rule). An important implication arises if the encoding occurs through a fixed, unchanging ‘channel’, $p_z(r|z)$, such that $p(r|x) = p_z(r|z = \varphi(x))$, with φ invertible, and where z describes for instance some internal, psychological scale. From this assumption follows the inequality³⁰:

$$\int_{D_x} \sqrt{I_x(x)} dx = \int_{\varphi(D_x)} \sqrt{I_z(z)} dz \leq \int_{D_z} \sqrt{I_z(z)} dz,$$

where D_x is the domain of x and D_z that of z . The last integral, assumed finite, is independent of φ . Thus even if one is free to choose the ‘transducer’ function, φ , the fixed-channel assumption puts a capacity constraint on feasible encoding schemes. Conceptually, the capacity constraint is that the total Fisher information across the stimulus space is bounded. This is the constraint assumed in Eq. 1.

Capacity constraint We have assumed that the encoding results from the accumulation of n samples. Each sample brings information about the stimulus, x , and helps minimize the expected loss; but we posit a limit on the one-sample information $I_1(x)$. Specifically, we assume the constraint

$$\int \sqrt{I_1(x)} dx \leq \sqrt{K}. \quad (1)$$

This constraint is used elsewhere^{25,26,28–30,47}; we explain in Box 1 why it is a natural one. Informally, it means that the encoding cannot be precise about all stimuli. It can be seen as a constraint on a ‘precision budget’ (notwithstanding the possibility of sampling several times, to which we turn below). For instance, if the encoding is carried by a single Poisson neuron with a monotonic tuning curve, this constraint is equivalent to assuming an upper bound on its firing rate.

Sampling cost We posit that each sample comes with a cost. This could represent, for instance, the energy required to fire action potentials^{48–50}; it can also be construed more theoretically as an opportunity cost corresponding to the loss incurred in all other tasks by employing resources in this particular one³⁵. Here we posit that the cost of one sample is λ , and thus the cost of n samples is λn . Unlike the capacity constraint, which governs how precision is distributed across stimuli, the sampling cost makes the overall precision of the representation an endogenous quantity that can adapt to the task and context. This assumption is crucial, and many of the results below depend critically on it.

Optimization problem and solution Putting together these ingredients yields an efficient-coding problem in which the observer chooses (i) how to allocate encoding precision across stimuli, $I_1(x)$, and (ii) how many samples to collect, n , so as to minimize the task’s expected loss under a sampling cost and a constraint on feasible encoding schemes. Formally,

$$\min_{n, I_1} L[nI_1] + \lambda n \quad \text{subject to} \quad \int \sqrt{I_1(x)} dx \leq \sqrt{K}. \quad (2)$$

Solving for n and I_1 , with task losses of the form $L_{f,m,p}[I]$, yields the optimal total Fisher information, $I(x) = nI_1(x)$, as⁵¹

$$I(x) = \frac{f(x)^{\gamma_{task}}}{C_{f,m,p}}, \quad \text{where} \quad \gamma_{task} = \frac{2m}{p+1}, \quad (3)$$

and where $C_{f,m,p}$ is a constant (see Supporting Information).

This shows, first, that the encoding should be more precise for more frequent stimuli: encoding resources ‘spread out’ across the stimulus space according to the relative frequencies of stimuli—a typical prediction of efficient coding. Second, the Fisher information depends not only on the prior but also on m and p , i.e., the representation depends on the task. Specifically, each task is characterized by a quantity, the exponent γ_{task} , which determines the allocation of resources, and thus the representation.

Box 2: A taxonomy of tasks and their losses

Different tasks imply different notions of error, and therefore different optimal representations. In the small-noise regime, the expected losses can be expressed as functionals of the Fisher information, $I(x)$.

Estimation tasks ($m = 1, p = 2$) A magnitude x is presented and the observer must estimate it. Here we use the term *estimation task* to refer to the case of squared-error loss. In the small-noise regime, the mean squared error is approximately the inverse Fisher information (Box 1). The expected loss is then

$$L_{est}[I] = \frac{1}{2} \int \frac{f(x)}{I(x)} dx. \quad (4)$$

Other p^{th} -power error metrics lead to the same structure but with $1/I(x)^{p/2}$.

Discrimination tasks ($m = 2$) These involve comparing two magnitudes. The relevant prior is the probability of drawing two nearby magnitudes ($x \approx y$), which is $f(x)f(y) \approx f(x)^2$. In *choice tasks* ($p = 2$), the reward equals the chosen magnitude (or is proportional to it). Errors are therefore more costly when the difference is large. In the small-noise regime,³⁰

$$L_{choice}[I] = \frac{1}{2} \int \frac{f(x)^2}{I(x)} dx. \quad (5)$$

In *classification tasks* ($p = 1$), a fixed reward is given for correctly identifying the greater magnitude. Unlike in choice tasks, confusing two nearly equal magnitudes is as costly as any other error. In this case³⁰,

$$L_{class}[I] = \int \frac{f(x)^2}{\sqrt{I(x)}} dx. \quad (6)$$

(See Supporting Information for derivations.)

A common family These cases can be unified, as³⁰

$$L_{f,m,p}[I] = \frac{1}{p} \int \frac{f(x)^m}{I(x)^{p/2}} dx. \quad (7)$$

This is the formulation that we use throughout (except for the change-localization task in the visual working memory literature, which implies a different loss function that we detail in Supporting Information). To each of these tasks is associated an exponent, $\gamma_{task} = 2m/(p+1)$ (see Eq. 3).

Mutual-information objective A common alternative in efficient-coding models is to maximize the mutual information between stimulus and representation. In the large- n limit⁵²

$$MI = \frac{1}{2} \int f(x) \ln I(x) dx + H(X) - \frac{1}{2} \ln(2\pi e) + o(1), \quad (8)$$

where $H(X)$ is the entropy. This corresponds to the limiting case

$$L_{MI}[I] = \lim_{p \rightarrow 0} \frac{1}{p} \int \frac{f(x)}{I(x)^{p/2}} dx, \quad (9)$$

i.e., $m = 1$ and $p \rightarrow 0$ within the same family.

Laws of psychophysics

Law #1: task-dependent sublinear scaling of imprecision

In everyday life, the same kind of magnitude may be encountered under different statistical contexts, corresponding to priors of different widths. How should the precision adapt when the range of likely stimuli expands or contracts? Because encoding resources are costly and must be allocated across stimuli, wider priors yield greater imprecision. But how the imprecision scales with the prior width depends on the observer’s objective, i.e., on the task. For uniform priors of width w , one obtains $I(x) \propto 1/w$ in estimation tasks and $I(x) \propto 1/w^{3/2}$ in choice tasks. Therefore the standard deviation of estimates in estimation tasks and the discrimination threshold in choice tasks verify, respectively, the following laws, first derived by Prat-Carrabin & Woodford⁵¹:

$$\sigma_{est} \propto \sqrt{w}, \quad (10)$$

$$\text{and } \sigma_{choice} \propto w^{3/4}. \quad (11)$$

These laws relate the scale of the internal imprecision to the scale of the prior. The scaling is sublinear and task-dependent: if the prior width doubles, the imprecision is multiplied by $\sqrt{2} \approx 1.41$ in estimation tasks, and by $2^{3/4} \approx 1.68$ in choice tasks. These theoretical predictions are verified quantitatively in numerosity estimation, motion-direction estimation, numerosity discrimination, and risky choice^{3,51,53,54} (Fig. 1a,b).

Important insights emerge from these results. First, representations adjust to changing priors, suggesting that efficient coding is dynamic rather than fixed to a long-run prior. This is consistent with evidence of prior sensitivity in perceptual tasks involving orientations, colors, shapes, speed, and numerosity^{47,76–80}. Second, representations depend on the task. This is consistent with evidence that sensory encoding is sensitive to reward structure^{81–84}, and more generally that task demands shape representations^{85–87}. Psychophysical measurements such as JNDs or biases are therefore not intrinsic, immutable characteristics of subjects, but can vary with the objective of the observer, and the JND measured in one task may not equal the JND measured in another task—for the same physical stimulus.

Third, perhaps counter-intuitively, the scaling is sublinear (by contrast, ‘range normalization’ models^{88–91} predict proportionality, $\sigma \propto w$). The reason is that the optimal number of signals, n , also grows with the prior width: specifically, $n \propto w$ in estimation tasks, and $n \propto \sqrt{w}$ in choice tasks (Supporting Information), thus broader ranges lead to more samples. Why should that be the case? As the prior widens and encoding resources are increasingly spread thin, errors get larger as well; to mitigate this, it is optimal to ‘pay’ more in encoding resources—i.e., to increase n . A related treatment of representational precision as a decision variable appears in a model of intertemporal choice⁹². Here we note that the resulting sublinear scaling means that representations are relatively more precise with wider priors. This implies, for instance, that the scale used to communicate a quantity may affect the precision with which it is perceived.

Law #2: task-dependent generalized Weber’s law

Above we considered how the encoding changes across priors of different sizes. We now turn to its variations across stimuli. We consider priors in the ‘shifted power law’ family, i.e., with density $f(x) \propto 1/(x + d)^\alpha$. For these priors we obtain (see Eq. 3) the generalized Weber’s law:

$$\sigma(x) \propto (x + d)^{\frac{1}{2}\alpha\gamma_{task}}. \quad (12)$$

In discrimination tasks, where σ is interpreted as the discrimination threshold, Weber’s law corresponds to a proportionality relationship ($\sigma(x) \propto x$). Its generalization as in Eq. 12 accounts very well for discrimination data across many different sensory modalities⁹³. Here we thus provide an efficient-coding rationale for Weber’s law (see also Refs. 80,94). Two further predictions follow: the scaling relationship is determined by the shape of the prior (via α and d), and by the task (via $\gamma_{task} = \frac{2m}{p+1}$). For several sensory modalities, studies have successfully modeled the prior as a generalized power law, and the encoding as a Fisher information proportional to a power of that prior^{27,31}, consistent with the model above over several orders of magnitude (Fig. 1c). We emphasize that the scaling of $\sigma(x)$ is not merely of the form predicted by Eq. 12; it is quantitatively determined by the prior over magnitudes, independently measured. A further implication of the theory is that if the prior changes, then the behavior may cease to obey Weber’s law. Consistent with this prediction, a recent study reports experimental violations of Weber’s law, following manipulations of the prior⁸⁰.

To our knowledge a direct test of the sensitivity to the task has not been conducted. A prediction is that in choice tasks ($\gamma_{choice} = 4/3$) the imprecision should increase with the magnitude faster than in estimation tasks ($\gamma_{est} = 2/3$). This calls for experiments directly comparing the two tasks, with the same sensory modality (and the same prior). In the studies just mentioned, the operating hypothesis is that the encoding maximizes the MI, corresponding to $\gamma_{MI} = 2$ (the exponent $\gamma = 1/2$, also tested, poorly fits the data²⁷). But the same exponent—thus the same allocation of precision, and thus the same perceptual behavior—is obtained in *classification tasks* (i.e., $\gamma_{class} = 2$; see Box 2). The data are thus consistent with both MI maximization and the minimization of a classification-task loss, since $\gamma_{MI} = \gamma_{class} = 2$ (experimental reports, unfortunately, rarely specify the actual reward structure). Disentangling these explanations would require manipulating experimentally the loss function and observing the resulting perceptual decisions. Some results mentioned above suggest that the encoding is task-sensitive^{51,80–84}, but more direct tests are warranted to ascertain whether representations are generally adaptive, or whether some sensory systems invariably maximize the MI, regardless of the context.

Bias of the Bayesian mean

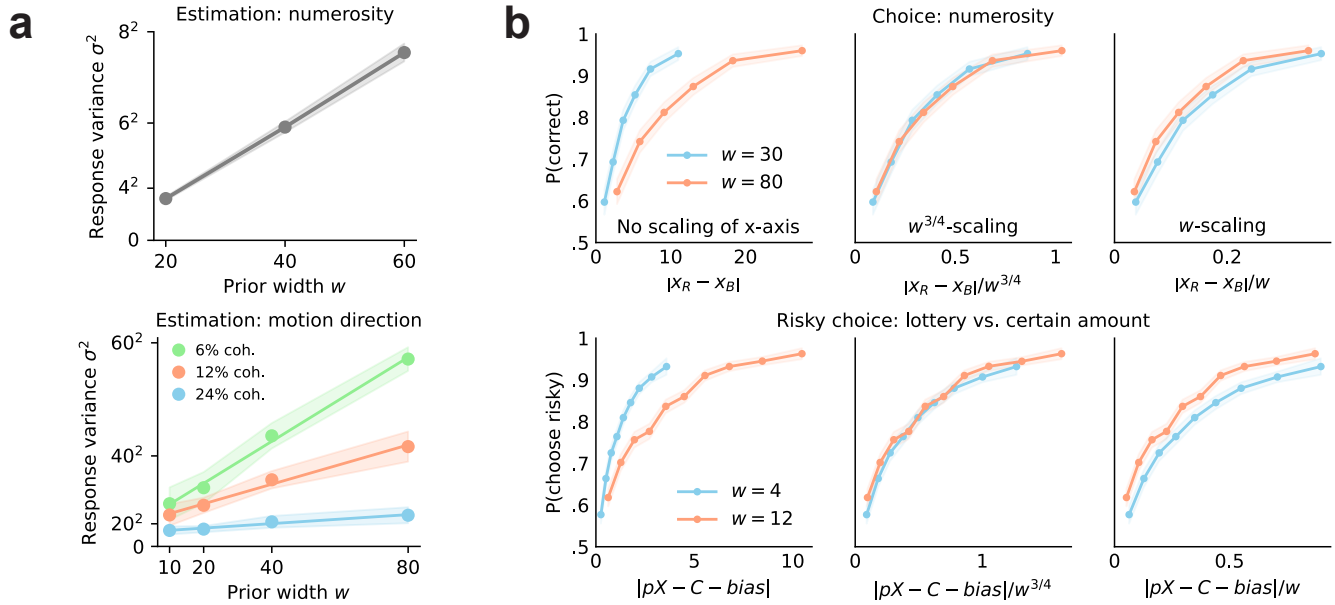
Imprecision generates variability, but also biases: a Bayesian decoder, typically, is biased by the prior toward more frequent stimuli, but it is also biased by the specifics of the

All these laws derive from a single equation: $\min_{n, I_1} L_{task}[nI_1] + \lambda n$ subject to $\int \sqrt{I_1(x)} dx \leq \sqrt{K}$.

Law #1: task-dependent sublinear scaling of imprecision

🎯 Estimation task: $\sigma \propto \sqrt{w}$

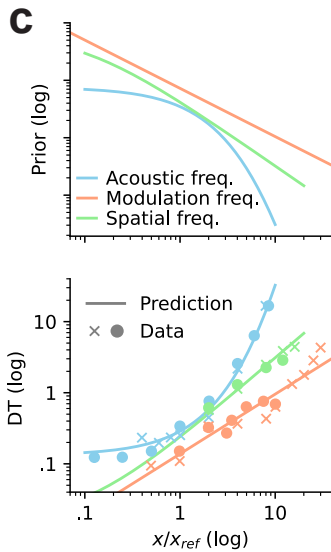
⚖️ Choice task: $\sigma \propto w^{3/4}$



Law #2:

Generalized Weber law

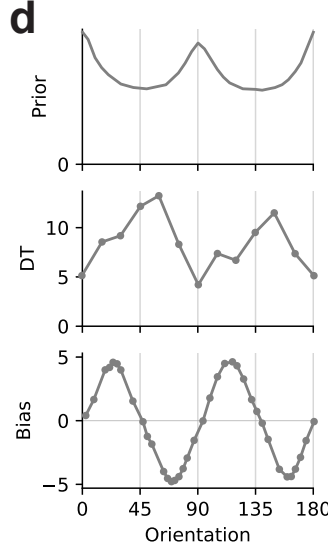
$$\sigma(x) \propto (x + d)^{\frac{1}{2} \alpha_{task}}$$



Law #3:

Wei-Stocker law

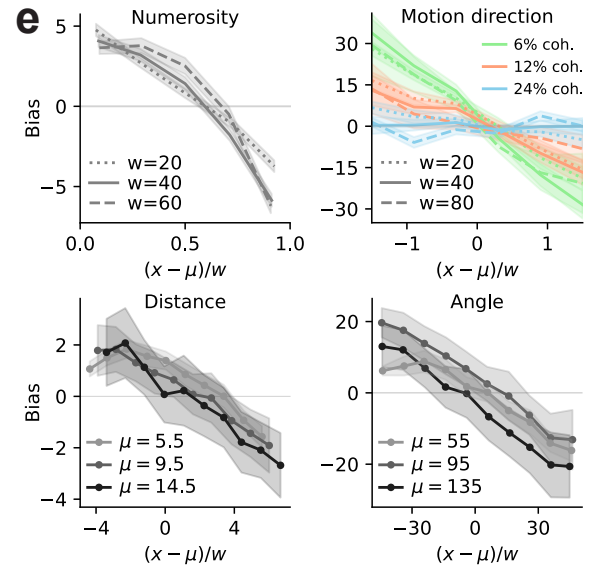
$$bias_{\gamma}(x) \propto \frac{d}{dx} \sigma_{\gamma'}(x)^{2\gamma/\gamma'}$$



Law #4:

Quantile invariance of bias

$$bias_{\mu, w}(x) = \beta\left(\frac{x - \mu}{w}\right)$$



Law #5: visual working memory:

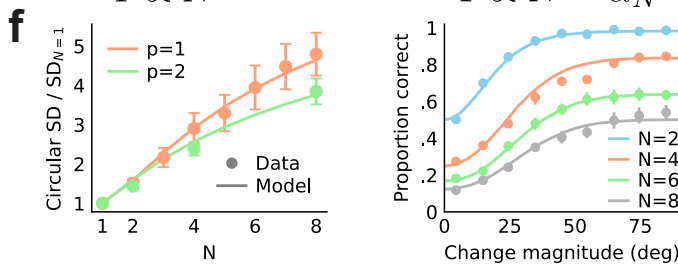
task-dependent scaling of precision with N

🎯 L_p -loss estimation:

$$I \propto N^{-\frac{4}{p+2}}$$

🔍 Change localization:

$$I \propto N^{-\frac{4}{3}} \alpha_N$$



Law #6: Shepard's universal law of generalization

$$G_{ij} = \exp(-d_{ij}^2)$$

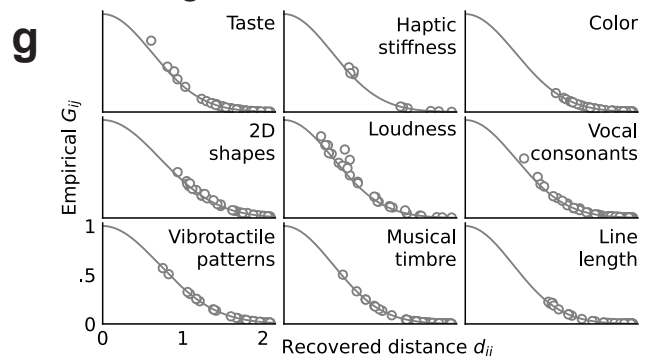


Fig. 1: Laws of psychophysics derived from endogenous efficient coding. (caption on next page)

Fig. 1: **a.** Response variance increases *linearly* with the prior width in a numerosity estimation task⁵¹ (top panel; shaded area: 5%-95% Bayesian credible interval) and in a motion direction estimation task⁵⁴ (random dot kinematograms with different levels of coherence; bottom panel; shaded area: sem). **b.** Choice probabilities under priors of different widths in a numerosity discrimination task⁵¹ (top row) and in a risky choice task³ (bottom row), as a function of the absolute difference between the two options (left column), of this difference divided by the prior width raised to the exponent 3/4 (middle column), and divided by the prior width (right column). The first column shows that participants are less precise with the wider priors, in absolute terms. The last column shows that, in *relative* terms, they are more precise with the wider priors. The middle column shows that their imprecision scales as the prior width to the power 3/4. (The three panels within a row show the same data with different scalings of the horizontal axis.) Shaded areas: 95% confidence intervals. **c.** Generalized Weber law: the prior (top panel) determines the discrimination threshold (bottom panel; predictions (solid lines) are derived from priors via Eq. 3 with $\gamma = 2$). Priors and data analysis from Ref. 27; data for acoustic frequency, auditory modulation frequency, and spatial frequency from Refs. 55–57 (crosses) and 58–60 (dots). **d.** Wei-Stocker law⁶¹: the discrimination threshold for visual orientations⁵⁷ (middle panel) increases where the bias⁶² (bottom panel) is positive, and it decreases where the bias is negative. The discrimination threshold moreover reflects the prior⁶³ (top panel) and is lowest for the more frequent orientations, consistent with efficient coding. **e.** Bias as a function of the centered, normalized stimulus $(x - \mu)/w$ in estimation tasks, with numerosity⁵¹ and motion direction⁵⁴ where the prior width, w , is manipulated (top row); and with distance and angle⁶⁴ where the prior center, μ , is manipulated (bottom row). Shaded areas: sem, except top left: 5%-95% Bayesian credible interval. **f.** Scaling laws in visual working memory tasks. *Left:* circular standard deviation (SD) in orientation-estimation tasks as a function of the number of presented items, N , relative to the circular SD at $N = 1$, and predictions of endogenous efficient coding with $p = 1$ (compared to data from 65) and $p = 2$ (data from 66). With $p = 1$ the circular SD increases faster than with $p = 2$. *Right:* Proportion of correct choices in a change-localization task⁶⁵ as a function of the magnitude of the change and per number of items, and model predictions. Error bars: sem. **g.** Shepard’s universal law of generalization. Generalization measure G_{ij} as a function of the recovered distances in psychological space, d_{ij} : data^{67–75} (circles) and predictions (lines).

encoding. Consider the Bayesian-mean estimator (which is optimal for estimation and choice tasks). In the small-noise regime, when a series of independent (but not necessarily identically-distributed) signals encode the stimulus x , the bias of the Bayesian mean, $\mathbb{E}[\hat{x} - x|x]$, is approximately³⁰

$$\text{bias}(x) \approx \frac{1}{I(x)} \left[\frac{f'(x)}{f(x)} - \frac{I'(x)}{I(x)} \right]. \quad (13)$$

A generalization exists for the optimal p -norm loss decoder, under assumptions on the encoding⁹⁵. The scaling term is the inverse Fisher information: more precise representations yield smaller biases. The term f'/f is a ‘prior effect’ which pulls estimates toward more frequent stimuli. Perhaps less intuitive, the ‘encoding effect’, $-I'/I$, pulls the Bayesian mean toward the less-precisely encoded stimuli. We illustrate this in Box 1.

As we have seen, if the encoding is optimally adapted to the prior and to the task, then the Fisher information is proportional to a power γ_{task} of the prior (Eq. 3), and thus the encoding effect is proportional, but opposite, to the prior effect, i.e., $-I'/I = -\gamma_{\text{task}}(f'/f)$. Depending on γ_{task} , i.e., on the specifics of the task, the encoding effect may dominate, thus accounting for ‘anti-Bayesian’ biases²⁵, directed toward the less probable stimuli. Furthermore, the bias depends on the variations of the Fisher information, and therefore on the variations of the imprecision; we now turn to this relationship.

Law #3: Wei & Stocker law relating bias and imprecision

The expressions of the bias (Eq. 13) and of the optimal Fisher information (Eq. 3) imply a relationship between the bias and the derivative of the imprecision. Before detailing this relationship, we note that for a given sensory modality, different behavioral measures may be determined with different tasks, e.g., biases with estimation tasks, and JNDs with discrimination tasks. But as noted above, these measures depend on the tasks themselves. Thus in order to make a precise statement about the relation between bias and imprecision, we consider the case in which the former, bias_γ , is measured in a task with exponent γ , while the latter, $\sigma_{\gamma'}$, is measured in a task with exponent γ' . Equations 3 and 13 imply:

$$\text{bias}_\gamma(x) \approx (\gamma - 1) \mathcal{C} \frac{d}{dx} \sigma_{\gamma'}(x)^{2\gamma/\gamma'}, \quad (14)$$

where $\mathcal{C}$ is a constant (see Supporting Information).

Wei & Stocker demonstrate that this relationship enjoys qualitative experimental support across many sensory modalities⁶¹. Specifically, they find that alternations in the sign of the bias coincide with alternations in the variations of the imprecision—which themselves coincide with alternations in the variations of the prior (Fig. 1d). They formulate a version of this law, $\text{bias}(x) \propto \frac{d}{dx} \sigma^2(x)$, which implicitly assumes that bias and imprecision are measured with the same task, although the results they report rely on many studies with different experimental protocols.

Empirically, the data suggest a positive coefficient for most modalities, though not all⁶¹. But if the encoding is optimized to a squared-error loss in the task measuring the bias (i.e.,

$\gamma = \gamma_{est} = 2/3$), then Eq. 14 predicts a negative coefficient instead. This suggests that participants may not minimize the squared error, at least in these studies. We note that in none of them is the reward tied to the estimation error, and in most cases participants receive no feedback (which would allow them to learn the objective of the task). Most of these tasks are matching tasks (‘method of adjustments’); recent work suggests that behavior in such tasks is better captured by an objective that includes a categorical-loss component, in addition to an error term⁹⁶. This may yield a different optimal allocation of precision (and thereby a different sign of the coefficient). Another possibility, as noted above, is that some low-level sensory systems implement a fixed objective, such as MI maximization, rather than adapting to the task. Distinguishing between these explanations will require carefully designed experiments in which the objective is explicitly manipulated and controlled.

Law #4: Quantile invariance of the bias

Many priors belong to location-scale families, parameterized by a location, μ , and a scale, w . We note a ‘quantile invariance’ property of the bias in this case: in estimation tasks, the bias is a function of the quantile of the magnitude, and not of the specific prior. Specifically, Eqs. 3 and 13 entail:

$$\text{bias}_{\mu,w}(x) \approx \beta\left(\frac{x-\mu}{w}\right), \quad (15)$$

where $\beta(\cdot)$ does not depend on the prior.

Consider two priors with different scales, $w \neq w'$. Quantile invariance implies that the bias for a magnitude $2w$ away from the mean, under the first prior, equals the bias for a magnitude $2w'$ away from the mean, under the second prior. This property appears verified in numerosity and orientation estimation data^{51,54} (Fig. 1e, top row). This invariance is not trivial, and follows from the endogenous adjustment of precision (modeled as the ability to choose the number of samples, n). If the imprecision increased linearly with the scale of the prior, the bias for a given quantile would also increase linearly; instead, the sublinear increase (Law #1) makes the bias invariant across priors.

Similarly, with priors that differ only in location ($\mu \neq \mu'$ but $w = w'$), the bias is predicted to depend only on the quantile (equivalently, on the relative location $x - \mu$). This is roughly consistent with magnitude estimation experiments that test ranges of equal size, but shifted⁹⁷ (Fig. 1e, bottom row).

Law #5: Task-dependent scaling of imprecision in visual working memory

So far we have examined how perceptual behavior varies with the stimulus, x , and with the parameters of the prior, e.g., w . Often, one has to represent several things at the same time. How are representational resources allocated as a function of the number of items, N ? Here we need to make an additional assumption, as our constraint (Eq. 1) does not say anything on how the one-sample Fisher information may be partitioned across several variables. From a model of a

network of sensory neurons with bell-shaped tuning curves, we find that the one-sample Fisher information naturally scales *quadratically* with the number of neurons in the population, provided that the network verifies a tiling property whereby the widths of tuning curves are inversely related to their density^{24,28} (see Supporting Information). Assuming that each variable is encoded by a subpopulation of a larger network whose total number of neurons is fixed, the number of neurons per variable decreases as $1/N$; therefore, the one-sample Fisher information for each variable decreases as $1/N^2$.

This quadratic decrease does not take into account the flexibility afforded by the ability to adjust the encoding precision via the choice of the number of samples, n . We thus solve the efficient-coding optimization problems that arise in tasks requiring the perception of the orientations of several items, as found in the literature on visual working memory^{65,66,98,99}. (This literature also uses color, but orientation provides a more unambiguously one-dimensional stimulus dimension.) In estimation tasks, where participants report their estimate of one of N orientations previously displayed, if the loss function is the p^{th} power of the estimation error, then the Fisher information for each variable is predicted to scale with N as (see Supporting Information)

$$I \propto N^{-\frac{4}{p+2}}. \quad (16)$$

In change-localization tasks, where participants report which of N items has changed across two successive displays, the Fisher information is predicted to scale as

$$I \propto N^{-\frac{4}{3}} \alpha_N, \quad (17)$$

where α_N is a term increasing sublinearly with N , related to the probability of misperceiving an unchanged item as having changed more than the correct item (this is not a free parameter; see Supporting Information, where we also derive the expected loss for this task, which falls outside of the family considered so far).

These predictions are verified in data. In an estimation task where the score at each trial was proportional to the estimation error⁶⁵, i.e., $p = 1$, the predicted scaling, $I \propto N^{-4/3}$, best fit the data (Fig. 1f, left; we follow the literature by implementing ‘variable-precision’ models in which the imprecision is itself stochastic^{65,66,99,100}; see Supporting Information). In another study which does not report that a loss function was specified nor that feedback was provided⁶⁶, the data is best fit by the assumption $p = 2$ (implying the scaling $I \propto N^{-1}$), suggesting that in the absence of instructions participants minimize squared errors (Fig. 1f, left). As for change-localization tasks, the specific scaling in Eq. 17 captures behavior very well across numbers of items and change magnitudes (Fig. 1f, right; data from⁶⁵).

Many models in the literature posit a power-law scaling (e.g., $I \propto N^{-2\alpha}$), where the exponent is a free parameter whose value is left unexplained^{65,66,98,99}. The efficient-coding model of van den Berg & Ma¹⁰¹ is closest to ours, with a cost linear in both the Fisher information and N —thus implying an endogenous choice of precision, as in our proposal—but where the loss function is the estimation error raised to an

exponent that is a free parameter, whereas we argue that the loss is determined by the task. Our argument, in short, is that by combining (i) endogenous precision, (ii) a careful analysis of the task’s objective, and (iii) our assumption, motivated by the tiling property, on how the one-sample Fisher information scales with N , one can identify the specific scaling of imprecision that should emerge in a given scenario. Overall, once again we find that representations are predicted to be task-sensitive, but here too more direct tests of this dependence are called for. We note that van den Berg *et al.*¹⁰² find that increasing monetary rewards does not improve visual working-memory precision, suggesting that precision may not respond strongly to all forms of incentive, but nevertheless depend on the structure and objective of the task.

Law #6: Shepard’s universal law of generalization

The framework makes interesting predictions even if the relevant underlying space is unclear—for instance, if it is a ‘psychological space’ unobservable to the experimenter. About 40 years ago, Shepard proposed that the probability of generalization, i.e., that the learned response to a stimulus generalizes to another, is a function of the distance between the two stimuli in some psychological space¹⁰³. We argue that this should indeed be the case for an optimal observer whose encoding is limited by the capacity constraint we have assumed (Eq. 1).

To see this, in the one-dimensional case, consider the monotonic transformation

$$z = \varphi(x) = \frac{\int_{-\infty}^x \sqrt{I(\tilde{x})} d\tilde{x}}{\int_{-\infty}^{\infty} \sqrt{I(\tilde{x})} d\tilde{x}} \in [0, 1]. \quad (18)$$

The reparameterization property of the Fisher information (Box 1) implies that the Fisher information with respect to z is constant: specifically, $I_z = K$ (if the constraint saturates). In other words, the precision in z -space is constant.

In the identification tasks that Shepard uses to illustrate his proposal, participants are asked to identify which of M possible items is presented to them. In z -space, the optimal estimator is asymptotically Gaussian with variance $1/K$, thus the probability of identifying z_i as z_j , denoted by p_{ij} , is approximately proportional to $\exp(-\frac{1}{2}K(z_i - z_j)^2)$. Shepard’s measure of generalization, defined as $G_{ij} = (\frac{p_{ij}p_{ji}}{p_{ii}p_{jj}})^{1/2}$, is then

$$G_{ij} = \exp(-\frac{1}{2}K(z_i - z_j)^2). \quad (19)$$

This suggests that there exists a space in which the generalization measure is a Gaussian function of the distance between two stimuli.

In the nine identification-task datasets^{67–75} gathered by Sims¹⁰⁴, where each dataset is an $M \times M$ confusion matrix, the data are consistent with this prediction (Fig. 1g). In other words, in each case the empirical matrix $-\log G_{ij}$ is indeed the matrix of the squared distances between M points in some space (how we recover these points and the corresponding distances is detailed in Supporting Information).

Several questions remain open. Shepard’s law is often written as $G_{ij} \propto e^{-d_{ij}}$, where d_{ij} is a distance recovered by a method called multidimensional scaling (a 2003 review, however, shows that Gaussian functions of the kind we obtain, $e^{-d_{ij}^2}$, typically better describe generalization data¹⁰⁵). Sims¹⁰⁴ proposes an efficient-coding model of generalization inspired by rate-distortion theory, and derives a solution of the kind $G_{ij} \propto e^{-\mathcal{L}(x_i, x_j)}$, where $\mathcal{L}$ is a loss function inferred from the confusion data. Shepard’s, Sims’s, and our accounts thus provide different but related mathematical descriptions of the same data, and it remains unclear which, if any, best captures the underlying representation and computation. The difficulty lies in identifying, from the confusion data, both a latent geometry (or a loss function), and the response rule mapping that latent quantity onto observed choices. Each account addresses this ambiguity by imposing different constraints; ours requires the exponent to be a squared distance in some space. We note however that any encoding satisfying the capacity constraint (Eq. 1) yields the constant-precision z -space constructed above and therefore, under the assumed Bayesian decoding, obeys the generalization law. However, because the relevant loss is unclear for many of the tasks—e.g., it is not obvious what loss should be assigned to confusing one vocal consonant with another—we cannot assess whether representations in these experiments are optimal in the efficient-coding sense. Our analysis therefore establishes a structural consequence of capacity-limited encoding, rather than the optimality of the particular representations.

Flexible neural representations

The laws just presented illustrate the range of behavioral regularities that follow from the endogenous efficient-coding theory; but it has important implications for neural coding as well. Notably it implies that the neural code is not static: when the prior changes, neural resources should reorganize accordingly. In the brain, most magnitudes are encoded by vast networks of neurons with bell-shaped tuning curves, each characterized by its ‘preferred stimulus’ (i.e., the stimulus to which a neuron maximally reacts). The hypothesis of *distributed range adaptation* posits that the distribution of neurons’ preferred stimuli flexibly ‘stretches’ to match the current prior (Fig. 2a,b). Consistent with this prediction, a recent fMRI study shows that when the prior changes, the receptive fields of numerosity-sensitive parietal populations collectively shift their preferred numerosities, thereby efficiently covering the stimulus space⁷⁹ (Fig. 2). These neural changes, moreover, are predictive of corresponding changes in response variability. A similar flexible coding has been reported for the representation of time in the rat’s striatal and hippocampal populations^{106,107}. Such adaptive coding may be implemented through context-dependent gain modulation in recurrent sensory networks⁵³.

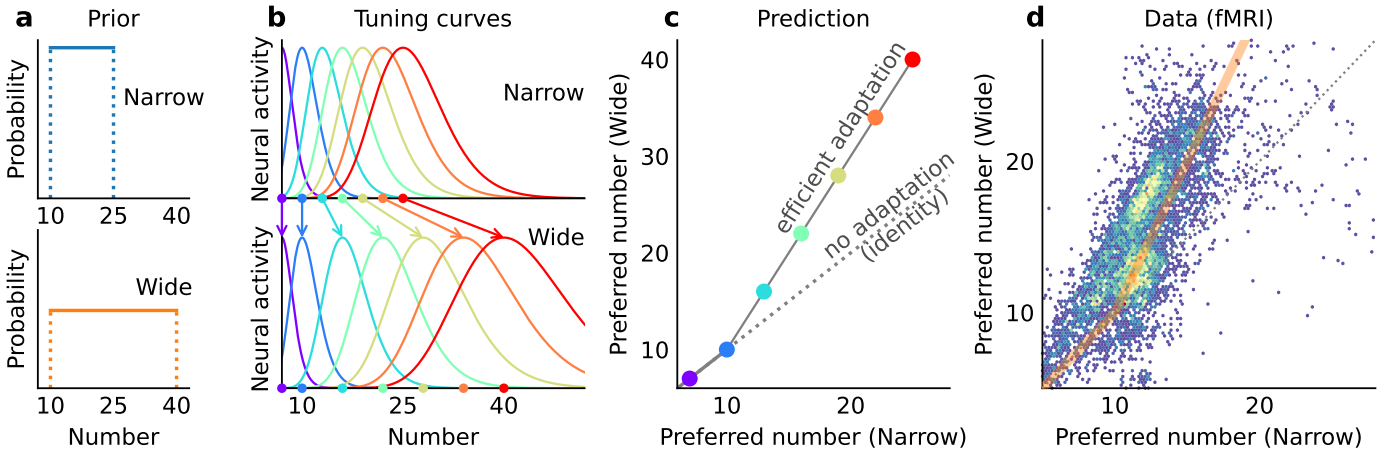


Fig. 2: Distributed range adaptation in neural tuning. Distributed range adaptation posits that when stimuli statistics change from a uniform prior to another (a), tuning curves collectively shift to efficiently cover the range of the current prior (b). This implies a linear relationship between the neurons’ preferred stimuli under the two priors (c), which was verified quantitatively in fMRI data collected in a numerosity estimation task⁷⁹ (d). This illustrates neural populations dynamically implementing efficient coding (Eq. 3).

Imprecision shapes behavior

The behavioral implications of representational imprecision are not limited to noise that averages out. As reasoning involves nonlinear transformations of information, imprecision generates deviations from optimality in complex ways. We illustrate three cases.

Primacy and recency effects These well-known effects can be explained by a single mechanism of Bayesian estimation. In sequential averaging tasks, participants estimate the average of a sequence of numbers. Assuming they maintain a running estimate of the average, they need to keep track of *the number of observations seen thus far* to determine how much weight to assign to the new number, in the updated average. If their memory is imperfect, they are only able to imprecisely estimate this count; typically, a Bayesian observer will overestimate small counts and underestimate larger ones (Fig. 3a, left panel). This alone generates primacy and recency effects in the weights assigned to the successive numbers, on the basis of these imprecise counts^{108,109} (Fig. 3a, right panel).

Risk attitude When offered a ‘safe’ (certain) option or a ‘risky’ one (a lottery), participants’ choices depend on whether the risky option is presented first or not. If it is, participants are more risk-seeking when values of options are low, and more risk-averse when they are high (Fig. 3b). This is readily explained if one assumes that the option presented first is less well remembered, and is thus estimated with a stronger ‘central tendency’, inflating lower values and depreciating higher ones⁷. More generally, many patterns of risk attitude are explained by models of cognitive imprecision, supported by neurophysiological evidence^{1-3,5-7,111}.

Inference Inferring the process that generates a sequence of observations enables making better predictions. With

sequences of Bernoulli signals, Bayesian beliefs converge, but the predictions of human participants fluctuate with the recent history of observations. Counter-intuitively, predictions in some cases are biased away from past observations (Fig. 3c), a repulsive effect captured by a model of inference under a cost on the precision of representations¹¹⁰. More generally, constraints on the computational resources required to represent beliefs and conduct Bayesian inference seem to underlie many seemingly suboptimal behavioral patterns in inference tasks¹¹²⁻¹¹⁹.

From representations to cognition

All cognitive models make assumptions about representations. Many posit precise representations and build elaborate suboptimal decision processes on top of them. Given that representations are necessarily imperfect, a sensible alternative is to consider instead optimal reasoning on the basis of imperfect representations. The examples above illustrate successful applications of this approach, where long-known behavioral patterns are readily explained by *optimal* decision-making over *imprecise* representations¹²⁰. Under this view, understanding cognition begins with identifying representations. We have argued that representations themselves are optimal, considering the context and resource constraints—with the immediate implication that representations are context-dependent. Explicitly identifying the constraints is crucial, and turns qualitative ideas into quantitative, testable predictions. Here we have done so by drawing on the literature on neural representations of magnitudes. The resulting theory is thus grounded in observable neural activity (Fig. 2), and has implications for both cognition and neural coding.

Important questions remain. Do early sensory representations adapt to task demands, or do they optimize a fixed objective such as mutual information, and where along the pro-

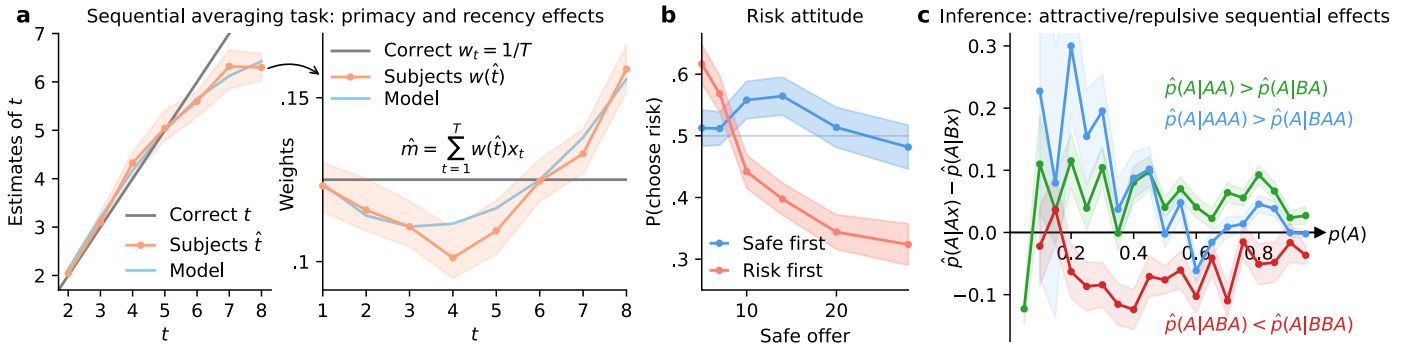


Fig. 3: From imprecision to bias. **a.** Imprecision in estimating the number of observations seen (left) yields primacy and recency effects (right) in a sequential averaging task^{108,109}. Shaded area: 95% confidence intervals. **b.** The order of presentation of safe and risk options impacts risky choices⁷. Shaded area: sem. **c.** Inference under a cost on the precision of beliefs results in subtle patterns of attractive and repulsive effects of past observations on current predictions¹¹⁰. Shaded area: sem.

cessing hierarchy does task dependence emerge? What does the number of samples, n , correspond to mechanistically—sampling duration, the size of the recruited neural population, or something else—and what predictions does this make for response times or behavior under time pressure? How are resources allocated when several multidimensional variables must be represented? How do these principles extend to more complex objects in richer spaces (e.g., semantic categories, relational knowledge)? And, ultimately, how does imprecision in such representations shape higher-level processes such as inference, learning, and memory? Combining first principles with empirical input from the neuroscience of perception provides a path toward a neurocognitive theory of representations, itself a foundation for a theory of human decision-making. We venture that this approach will remain fruitful as it is extended beyond simple magnitudes to richer forms of information.

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